



$$a^2 - ax - 2x^2 - 6a + 3x + 9x = 0$$

$$-2x^2 + x(12 - a) + a^2 - 6a = 0$$

$$-2 \left(x^2 + x \frac{(12-a)}{2} + \left(\frac{12-a}{4} \right)^2 \right) + \left(\frac{12-a}{4} \right)^2 + a^2 - 6a$$

$$-16 \left(x + \frac{12-a}{4} \right)^2 + (12-a)^2 + a^2 \cdot 8 - 48a$$

$$144 + 24a + a^2$$

$$9a^2 - 42a + 144$$

$$9(a^2 - 14a + 16)$$

$$9(a-4)^2$$

$$-16 \left(x + \frac{12-a}{4} \right)^2 + 9(a-4)^2 = 0$$

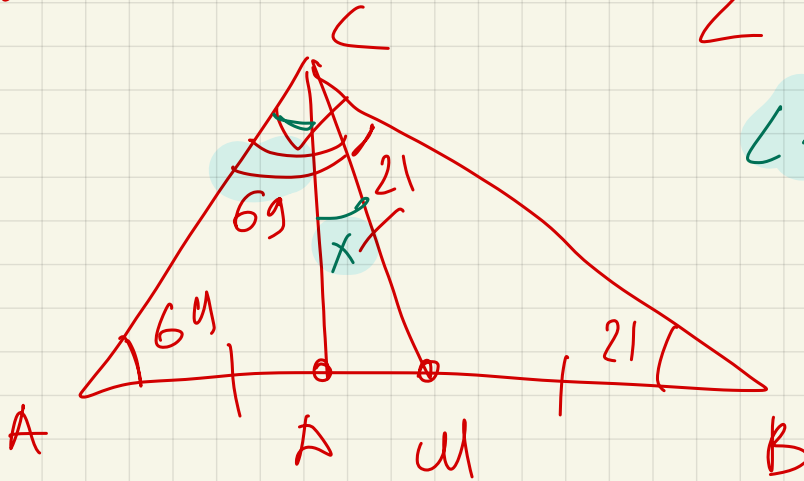
$$(3(a-4))^2 - (4 \left(x + \frac{12-a}{4} \right))^2 = 0$$

$$(3(a-4))^2 - (4x + 12 - a)^2 = 0$$

$$(3a - 12 - (4x + 12 - a))(3a - 12 + (4x + 12 - a)) = 0$$

~~2~~, ~~5~~, 7, 10

1, 8



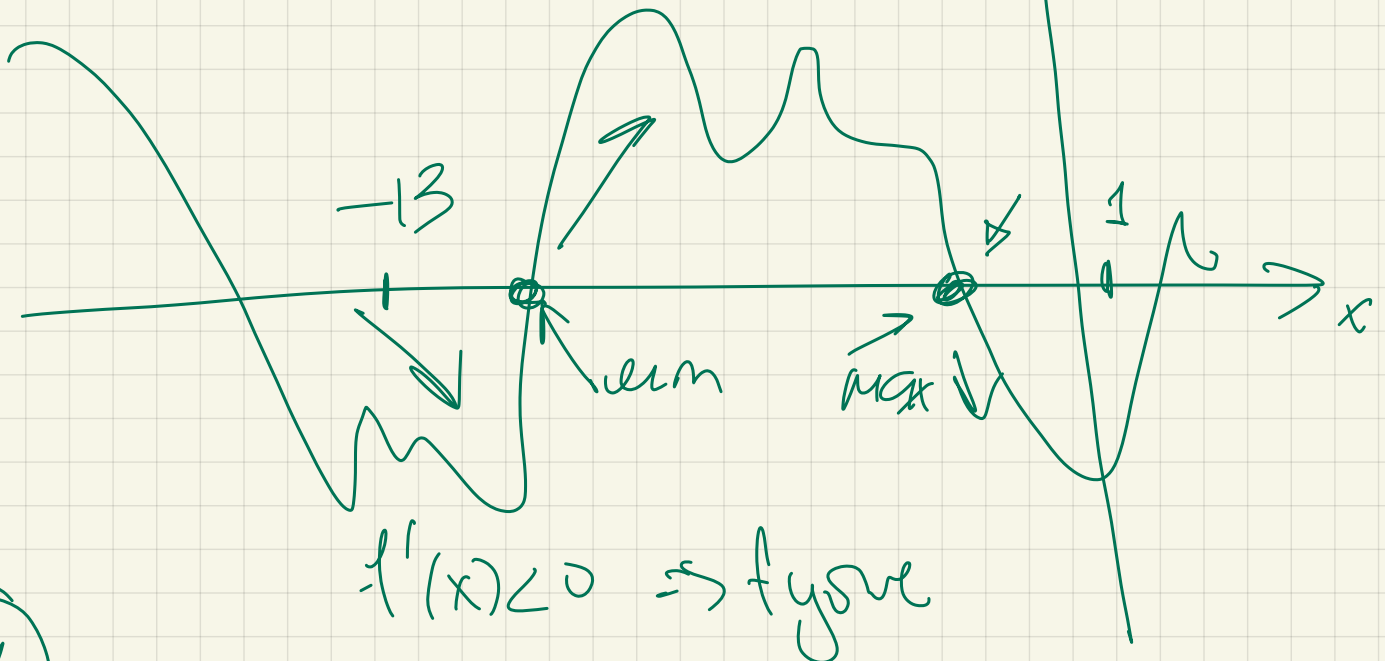
$$\angle ACM = 69$$

$$\angle ACU = 45^\circ$$

$$x + \angle ACP = \angle ACM$$

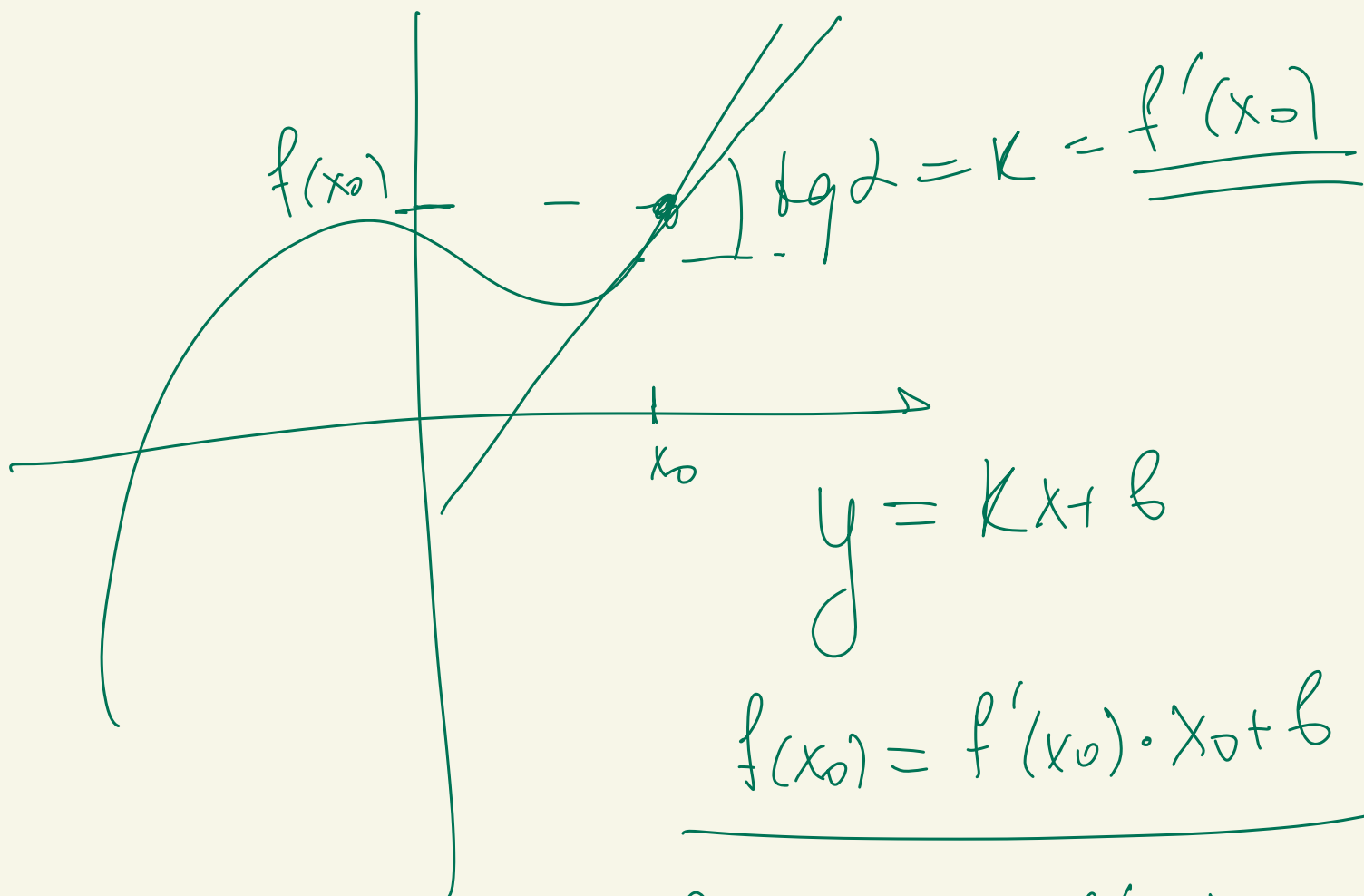
$$x = 69 - 45 = 24$$

$$\underline{\underline{f'(x) = 0 \Rightarrow f \text{ экстрем.}}}$$



$$f'(x) < 0 \Rightarrow f \text{ yvse}$$

1



$$\underline{\underline{b = f(x_0) - f'(x_0) \cdot x_0}}$$

$$y = f'(x_0) \cdot x + f(x_0) - f'(x_0) \cdot x_0$$

$$= f'(x_0 - x) + f(x_0)$$

$$y = -2x - 6$$

$$\underline{\underline{y = x^2 + 5x - 4}}$$

$$f'(x_0) = -3$$

$$y' = 2x + 5$$

$$y'(x_0) = -3$$

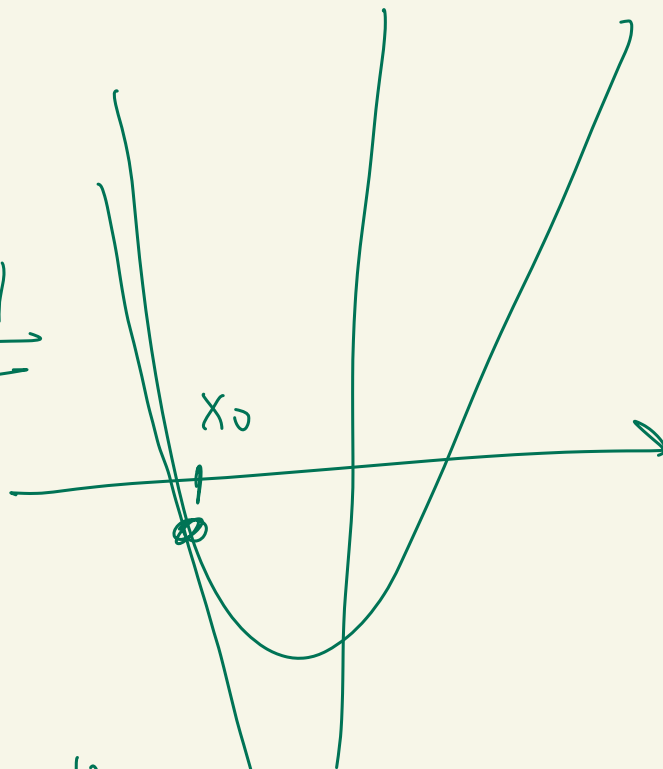
$$2x + 5 = -3$$

$$2x = -8$$

$$x = -4$$

$$y = \underline{\underline{-5x - 7}}$$

$$f(x) = \underline{\underline{8x^2 + bx + 11}}$$



$$\left\{ \begin{array}{l} f'(x) = -5 \end{array} \right.$$

$$-5x - 7 = 8x^2 + bx + 11$$

$$\left\{ \begin{array}{l} 16x + \underline{b} = -5 \end{array} \right.$$

$$b = -5 - 16x$$

$$-5x - 7 = 8x^2 + bx + 11$$

$$-5x - 7 = 8x^2 + (-5 - 16x)x + 11$$

$$-5x - 7 = 8x^2 - 5x - 16x^2 + 11$$

$$\cancel{-5x - 7} = -8x^2 - \cancel{5x} + 11$$

$$8x^2 = 11 + 7$$

$$x^2 = \frac{18}{8} = \frac{9}{4}$$

$$x = \pm \frac{3}{2}$$

$$x > 0$$

$$x = \frac{3}{2} \quad b = -5 - 16 \cdot \left(\frac{3}{2}\right) =$$

$$-5 - 24 = \underline{\underline{-29}}$$

$$y = 3x + 1$$

$$f(x) = ax^2 + 2x + 3$$

a?

$$\begin{cases} 2ax + 2 = 3 \\ 3x + 1 = ax^2 + 2x + 3 \end{cases}$$

$$a = \frac{1}{2x}$$

$$y = -5x + 8$$

$$y = 28x^2 + 6x + 15$$

$$b? \quad x > 0$$

$$y = -4x - 11$$

$$y = x^3 + 7x^2 + 7x - 6$$

найти x_0 ?

$$3x^2 + 14x + 7 = -4$$

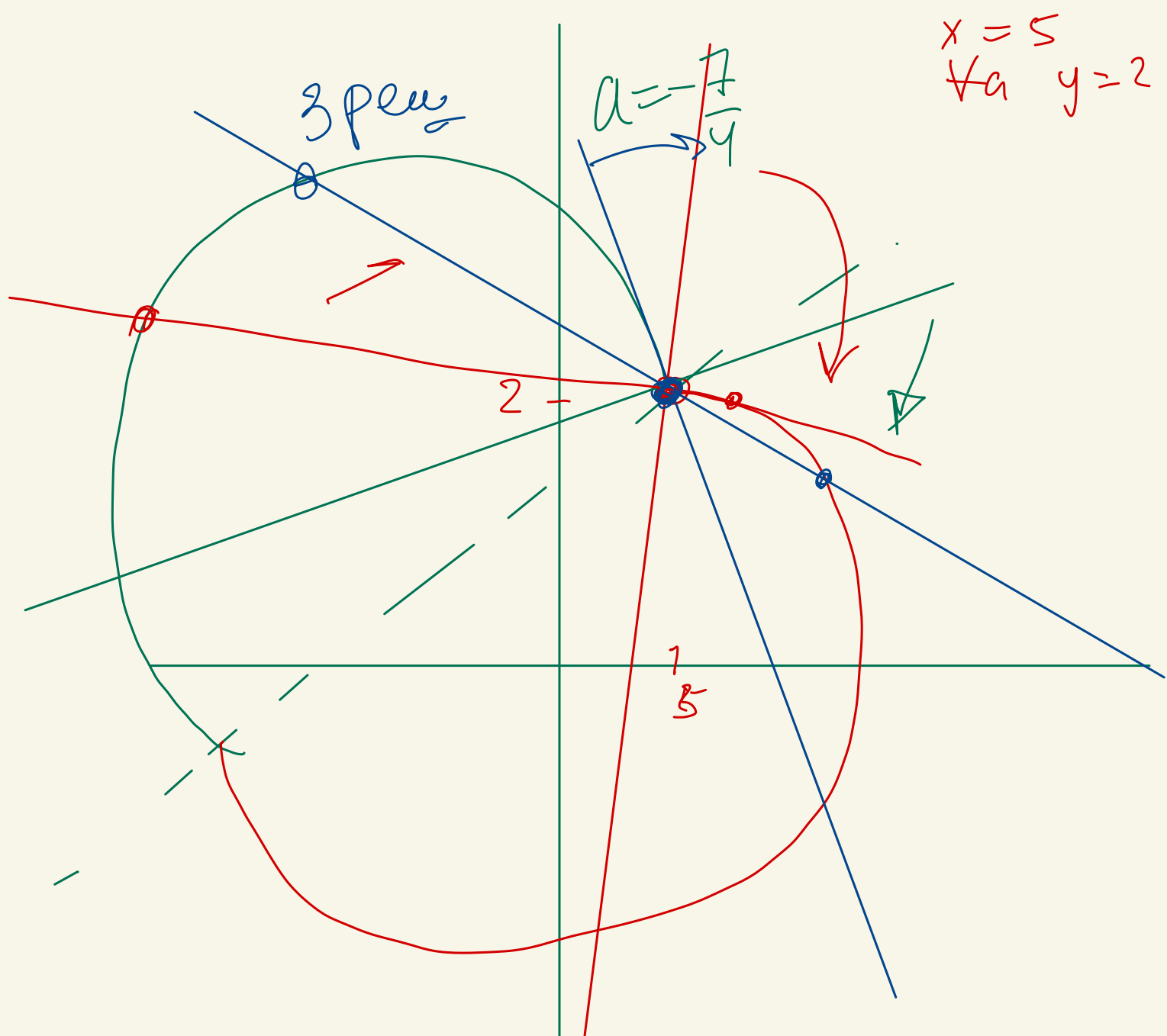
$$3x^2 + 14x + 11 = 0$$

$$\boxed{x = -1}$$

$$x = -\frac{24}{6}$$

$$y = ax - 5a + 2$$

$$a(x - 5) + 2$$



$$y = a(x-5) + 2$$

$$(x+2)^2 + (y+2)^2 = 65$$

$$D = 0$$

$$\left(a(x-5) + 4 \right)^2$$

$$x^2 + 4x + 4 + a^2(x-5)^2 + 16 + 8a(x-5) = 65$$

$$x^2 + 4x + 4 + \underline{a^2 x^2} + \underline{a^2 \cdot 25} - \underline{10x \cdot a^2} + 16 + \underline{8ax} - \underline{40a} - 65 = 0$$

$$\underline{x^2(1+a^2)} + x(\underline{4-10a^2+8a}) - 45 + a^2 \cdot 25 - 40a = 0$$

$$\Delta = 0 \quad \underline{(-10a^2 + 8a + 4)^2} - \underline{4(1+a^2)(a^2 \cdot 25 - 40a - 45)} = 0$$

$$(a+b+c)(a+b+c)$$

$$-90a^3 + \underline{224a^2} + \underline{224a} + 196$$

$64a^2$

$$\begin{aligned} &(-10a^2 + 8a + 4)(-10a^2 + 8a + 4) = \\ &\underline{100a^4} - \underline{80a^3} - \underline{40a^2} - \underline{80a^3} + 64a^2 + \underline{32a} \\ &\quad - \underline{40a^2} + \underline{32a} + \underline{16} \end{aligned}$$

$$100a^4 - 160a^3 - 16a^2 + 64a + 16$$

$$(1+a^2)(25a^2 - 40a - 45) = \underline{25a^2} - \underline{40a} - 45$$

$$+ 25a^4 - 40a^3 - \underline{45a^2}$$

$$4(25a^4 - 40a^3 - 20a^2 - 40a - 45) =$$

$$\underline{100a^4 - 160a^3 - 80a^2 - 160a - 4 \cdot 45}$$

$$\cancel{100a^4} - \cancel{160a^3} - 16a^2 + \underline{64a} + 16 - \cancel{100a^4} + \cancel{160a^3}$$

$$+ 80a^2 + \underline{160a} + 4 \cdot 45$$

$$64a^2 + 224a + 196 = 0$$

$$4(16a^2 + 56a + 49)$$

$$4(4a + 7)^2 = 0$$

$$a = -\frac{7}{4}$$

$$\Delta/3$$

$$x < 0$$

$$a^2 - ax - 2x^2 - 6a + 3x + 9|x| = 0$$

$$x > 0 \quad \underline{a^2 - ax - 2x^2 - 6a + 3x + 9x} = 0$$

$$x^2 + y^2 = R^2$$

$$(-2x^2 + x(12-a)) + a^2 - 6a = 0$$

$$\underline{-2} \left(x^2 - \frac{x(12-a)}{2} + \left(\frac{(12-a)}{2 \cdot 2} \right)^2 \right)$$

$$a^2 - \underline{(2)}ab + b^2$$

$$-2 \left(x - \frac{(12-a)}{4} \right)^2 + 2 \left(\frac{(12-a)}{4} \right)^2 + a^2 - 6a = 0$$

$$-2 \left(x - \frac{(12-a)}{4} \right)^2 + \cancel{2} \left(\frac{144 - a \cdot 24 + a^2}{\cancel{8} \cancel{46}} \right) + a^2 - 6a = 0$$

$$-16 \left(x - \frac{(12-a)}{4} \right)^2 + 144 - 24a + a^2 + 8a^2 - 48a = 0$$

$$-16\left(x - \left(\frac{12-a}{4}\right)\right)^2 + \frac{9a^2 - 72a + 144}{9(a^2 - 8a + 16)} = 0$$

$$\underline{-16\left(x - \left(\frac{12-a}{4}\right)\right)^2} + \underline{9(a-4)^2} = 0$$

$$-\left(4\left(x - \left(\frac{12-a}{4}\right)\right)\right)^2 + (3(a-4))^2 = 0$$

$$-\underbrace{(4x - 12 + a)^2} + \underbrace{(3(a-4))^2} = 0$$

$$-A^2 + B^2 = 0$$

$$(B-A)(B+A) = 0$$

$$(3(a-4) - (4x-12+a))(3(a-4) + (4x-12+a)) = 0$$

$$(2a - 4x)(4a - 24 + 4x) = 0$$

$$\left[\begin{array}{l} 2a = 4x \Rightarrow a = 2x \\ 4a - 24 + 4x = 0 \Rightarrow a = \frac{-4x + 24}{4} = -x + 6 \end{array} \right.$$

$$\left\{ \begin{array}{l} a = 2x \\ a = -x + 6 \\ x \geq 0 \end{array} \right.$$

$$a^2 - ax - 2x^2 - 6a + 3x - 9x = 0$$

$-6x$

$$-2x^2 - x(6+a) + a^2 - 6a = 0$$

$$= -2 \left(x^2 + x \frac{(6+a)}{2} + \left(\frac{6+a}{4} \right)^2 \right) + 2 \left(\frac{6+a}{4} \right)^2$$

$$+ a^2 - 6a = 0$$

$$-2 \left(x + \frac{6+a}{2} \right)^2 + 2 \left(\frac{6+a}{4} \right)^2 + a^2 - 6a = 0$$

$$-2 \left(x + \left(\frac{6+a}{a} \right) \right)^2 + \frac{36+12a+a^2}{8} + a^2 - 6a = 0$$

$$-16 \left(x + \left(\frac{6+a}{a} \right) \right)^2 + 36+12a+a^2 + 8a^2 - 48a = 0$$

$$9a^2 - 36a + 36 = 0$$

$$9(a^2 - 4a + 4) = 0$$

$$-16 \left(x + \left(\frac{6+a}{a} \right) \right)^2 + 9(a-4)^2 = 0$$

$$- \left(4 \left(x + \left(\frac{6+a}{a} \right) \right) \right)^2 + (3(a-4))^2$$

$$- (4x + 6 + a)^2 + (3(a-4))^2 = 0$$

$$- B^2 + A^2 = 0$$

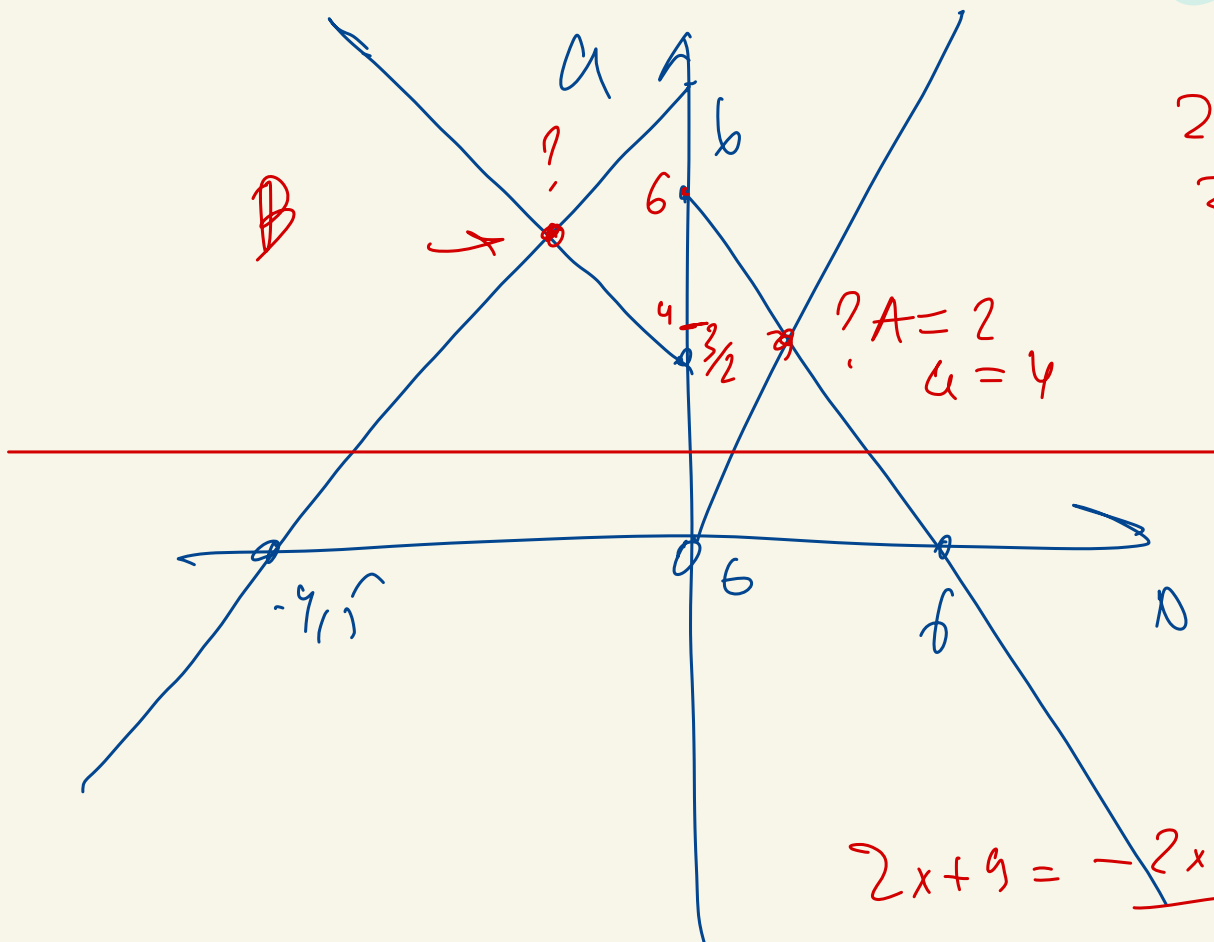
$$(3(a-4) - (4x + 6 + a))(3(a-4) + 4x + 6 + a)$$

$$(3a - 12 - 4x - 6 - a)(3a - 12 + 4x + 6 + a) = 0$$

$$(2a - 18 - 4x)(4a + 4x - 6) = 0$$

$$\begin{cases} 2a - 18 - 4x = 0 \\ 4a + 4x - 6 = 0 \end{cases}$$

$$\begin{cases} a = \frac{4x + 18}{2} = 2x + 9 \\ a = -\frac{4x - 6}{4} = -\frac{2x - 3}{2} \end{cases}$$



$$\begin{aligned} 2x &= -x + 6 \\ 3x &= 6 \\ x &= 2 \end{aligned}$$

$$2x + 9 = -\frac{2x - 3}{2}$$

$$n \in \left[\frac{3}{2}; 4 \right) \cup (4; 6) \cup (12; 6]$$

$$4x + 18 = -2x + 3$$

$$6x = -15$$

$$x = -\frac{15}{6}$$

$$a = 2\left(-\frac{15}{6}\right) + 9$$